

IOUBI Concepts repo:

Money is complicated?

We all know the symptoms.

We've all felt it. And recently, we can't avoid it anywhere. The cost of everything is rising. Incomes are falling for the majority. Quality and quantity are decreasing. More things cost that used to be free. More wealth is rising to the top and leaving the masses. Every institution is becoming increasingly corrupt. Money attracts more money faster. The lack of money, starves opportunities.

Why is this happening?

The global economy is migrating from producing actual physical goods and services, to mostly earning high returns on the ability to use money for pure profit. Car companies no longer just build and sell cars - they profit more from financing the loans. Banks don't just earn interest on loans - they upsell financial products that include hidden fees for every misstep in a complicated array of rules. From food to pharmaceuticals to insurance to transportation to homes, every product sale is being optimized to extract maximum profit. Every company is playing this game. Every individual is playing. Every government. Every charity. As they are forced to pay more to maintain their standing, they pass those costs on. And it doesn't stop there.

What is the result?

This competition to gain more from every interaction with everyone else has become the defining nature of society. But it has follow-on effects that make it worse. Money now translates to power because power is needed to alter the rules. This allows markets to be manipulated in creative new ways that the masses rarely see. The competition among the wealthy is no longer just for survival or for buying luxuries to show off with. Their game is to gain enough wealth that they can win in a new competition. When they want to change a rule, a law, a market or some public opinion, they must compete with other wealthy groups that want the opposite. Think of this like asking who is willing to spend the most to get their candidate to win a primary election. Think who can get their ideology most talked about by the media. These competitions set national agendas. They decide what the public ends up thinking of the economy's health or what they fear in their lives. The result is that no level of wealth is ever enough.

What are the biggest ways this happens?

As this game has grown, every way that can be imagined has been maximized for profit. Everyone is a player, either for survival or competition and everyone must play. The only difference is the ways they do, differ among different groups. The poor work harder or more jobs or resort to despair comforts or even crime when that fails. The working class must also protect their future by growing their savings to beat inflation. The professional and political class has yet another added burden of managing their fragile relationship with the rule makers. And the elite class must constantly avoid threats to their standing and power. Each is focused on problems at their level and partially ignorant of those of higher levels. No group is solely to

blame regardless of how the other groups feel. The structure of the system forces them all to do exactly what they do. They are trapped in their own cages that have replaced chairs in the new game of musical cages.

What structural feature causes this?

We're all mostly aware why this happens. Some call it inflation while others refine that cause into why inflation happens. There are many factors to pursue that economists rarely agree on most fundamentals. But all those factors are still just symptoms caused by the root problem. The way money works is not matching how it should.

What is money?

Money is the receipt of an IOU. That's it. It is a placeholder to track the mismatch of either quantity or time when trading for something. Without it, we would try to trade like we've learned that barter works. But true barter doesn't and has never worked. If a trade of cows for eggs doesn't match in both the total value and the timing, something must track the imbalance that remains. This results in credit - a ledger of who still owes what. If the cow was known to be worth 83 dozen eggs but only traded for one dozen today, the ledger must show one party still owes the cow's seller 82 dozen eggs. But neither party cares if they settle this with the other. They are fine with settling with anyone in society for what they deserve. The cow seller can show their receipt to someone else that might owe a pig, and collect that amount instead. To make this work, a standard unit is needed to set the value left over after every trade. This allows cows, eggs and pigs to be standardized to one type of unit. And ignoring history, we all know the world's most common standard has finally become the US dollar.

What is a dollar, really?

Dollars have a very rigid value, or so most people think. But it does vary and because this allows systemic manipulation, this causes various groups to argue over why this happens and how to fix it. But none of that really matters because the actual physical goods and services that we want to trade, are actually measured in another unit - . To produce anything, or even to alter it for profit, the time of human effort involved is the ultimate root standard of the measure of value. Some might say that resources like oil or gold are better because they are limited, scarce, stable or common. Others feel it's better to allow governments to just decree its value. But all current and past forms of money ultimately depend on the trust of that standard. So, if money is simply trusted and ultimately a measure of time, then we should use time as the standard instead. Right?

Time currencies are the answer?

Various groups over history have tried to use time as a currency standard. Each has failed to become the global standard but the reasons vary. Some remain local, others have no way of accounting for the difference between a mechanic's hour vs a brain surgeon's. Still others simply couldn't scale outside a local region or became subject to hoarding or tax payment issues. Those are real problems but they don't mean the concept fails. Their implementation

was flawed. What they all needed was an individual or group or region or national conversion between time and the actual currency unit used.

The currency standard is a personal conversion?

The missing piece in this is that the currency unit must stay stable to some global standard amount of time but each person's conversion of that amount depends on what they value their time as. This delta unit remains fixed between it and the product but varies between the unit and the person. A mechanic may value their time as each unit being worth 10 minutes while a brain surgeon may feel theirs is worth 2 seconds. Both will pay 2 units for a loaf of bread but they spent vary different amounts of time to earn that delta dollar. Their time value has become a personal decision and their delta dollars remain matched to the global standard.

So what sets the global Delta-Dollar's value?

The new currency unit, this Deltar, can be set by averaging the total spending of it in the world by the total time spent earning them. That would give an exact value and it still might change some. But a much easier way is to simply average what everyone chooses their own conversion rate to be. A global average of this would include the averages of everyone's collective psychological wants, needs, fears, hope in their personal decisions. They would all have their own list of reasons but the global average would settle a more stable, and reliable value than any other currency in human history has ever done. It may not be a hard math formula and it may not start out fully stable but it works much better than any math could.

How would this work in practice?

When buying something, that product or service presents a price based on the globally trusted Deltar's value. When people compare an apple's price between NW USA and a struggling village in Africa, they can now see it always staying at 1 Deltar. Each will feel this costs them more or less time but the labeled prices match. This is helpful to assess inflation. If the prices don't match, it then shows them that both apples required differing amounts of real labor - possibly because one party is using automation or not valuing their time correctly. These are valuable price signals that are hidden by the inflation in the current system. But over time, people will learn to read and apply them, to help balance global trade.

Inflation is complicated?

Inflation has 2 basic causes and many subsidiary factors. The most cited one is monetary inflation, caused by an increasing ratio between an abundant supply of money which people are using to purchase a falling quantity of products. This happens from changes in the supply/demand balance of both the products and the currency quantity. It can be isolated to a product, region, nation or it can be a global result of many different factors. The other main one is created by the sellers artificially manipulating their supply or their prices to increase their profits. They do this because of what many justifiably call greed but the root cause is their profit motive.

Is greed unavoidable?

People often argue that greed is a trait that is built directly into human nature. Whether this is true or not, we do see that greed for something diminishes when that item becomes abundant. Water used to be seen as being as abundant as grass clippings. We now know that it is not fully abundant and the market has capitalized on this to make it even more scarce and expensive. Grass clippings are abundant, given no useful demand so that cannot yet be an item that can extract value from us. When items get disrupted by technology, like photos in digital cameras or like mainframe computers did from PCs and phones, those products lose the appeal to be hoarded.

Is monetary inflation unavoidable?

It's well known that money supplies can be and have been altered throughout history. Regardless of the mechanism or controlling entity, the process itself is easy. To avoid inflation or its companion the deflation spiral, the exact supply quantity would need to be managed in a controlled way that eliminates all adverse effects. It would need to instantly create all the money required to match every outstanding IOU to society with no debt based cost. It needs to be exchanged for the exact value of each sale. Finally, it needs to be destroyed when any amount of debt is repaid back to society. The IOUs can be traded around the world without needing to be tracked but the system needs to ensure the original borrower cancels their outstanding IOUs before accumulating other ones. In other words, the ledger must track each human's lifetime net balance.

Currency volume equals debt volume

When currency is only created to match new debt, the two must match in quantity and value. Many people will have a net balance that's positive, as revealed by their account balance. Others will have net negative balances, indicating that they owe society an amount that contributes to the global supply of Deltars. If it should happen that everyone owing debt was able to repay the ones holding the Deltars by selling things, the money supply would vanish completely, indicating a 100% balanced economy.

How do we track big loans?

For loans of every type, we can no longer recognize that they all work identically - the numbers are just larger. Most people, however, miss seeing that selling stocks in a company is just a loan. At its most fundamental level, selling shares of stock, like selling a bond, is just a complicated loan. The company creates an IOU (money) and trades it for other money. The idea that it is a part of the ownership of the company is common because shareholders often have decision power in company actions but that's identical to how some loans work. If the company was to repay that loan by buying the stock back, that debt has been repaid and the shares destroyed. And even if other financial instruments are created based on some shares, that's still just buying and selling one form of money for another. As long as the company can simply overdraft all they wanted to borrow at no interest cost, all that activity is a needless and expensive process. The overdrafted money will always be better, free and faster, giving a massive advantage to businesses that go this way.

What limits this free money?

The idea that loans are free and loans are unlimited are completely separate. Free means they cost nothing to obtain. Unlimited means their loan amount can grow to any number. Free is good, as it stops the exponential growth mandate on the money supply. But that supply of money needs to be matched exactly to the demand for IOUs for all the things that people want to buy. This doesn't mean that everyone gets enough to buy yachts and helicopters for their mansions. It means that the collective goal of the entire planet's population gets a voice in that vote. Assuming that everyone's living standards are equal, everyone earned the same and staples were more abundant than luxuries, the world average spending would show how much total it takes to be average. Items that actually were more scarce would be bought less (due to supply constraints that cause demand to fall), because people would stop buying that item when the price rose too much. Because of this, we can simply let everyone create that average and then assign that as their credit limit. With this done across the board, we now adjust it up and down some for regional differences in living costs and for healthy vs bad economic behavior.

Is this a fixed credit limit?

The limit that would result would change but slowly. It will always follow the global averages of the Deltar's purchasing power, the cost/benefit comparison of each item by the people and the actual scarcity presented by the real supply chains. If the trend was that the wealthiest began gaining wealth faster than the majority, everyone's credit limits would increase. This only helps the poor to catch up since credit is only useful to overdraft when someone is short on money.

What about those with emergencies?

Since IOUBI is a digital, global and automated system, we can implement a small fee on every transaction that covers every societal cost at once. Examples of these 'commons' costs include the collective cost of governments, grants, insurance and helping the disabled survive. All people would have identical opportunities to thrive from the free credit with identical safety net

support for when luck turned against them. The cost would be spread across every transaction in the entire system, making it almost unnoticeable, around 1/4th of a percent.

We don't know if problems will get better or worse.

The debate rages on as to whether paid jobs will increase or decrease due to automation and AI. Most opinions suggesting this will be bad all stem from the fear of starving or at least losing ground against inflation. Others just want to ensure they get fair credit for their creations, regardless of the field. Those suggesting it would be wonderful, feel it will make everyone rich. The reality is that since people are messy, the AI question will likely follow how people use it. In a chaotic system of mandate profit as the top goal, AI will accelerate that. In a system where money tracks people's time owed to society, acting as a measure of productive activity, AI will accelerate that too.

Not every poor person was the result of an emergency.

For those just getting poorer by the structure of the economy, we need to increase their income floor so they can at least survive. This shouldn't be just a minimum amount though because we may all be automated out of a job at some point. It should be at the level that offers a minimum standard of living to be fully productive, healthy and happy. Most luxuries shouldn't qualify but as times change, that definition changes too. By changing that fee from ¼ percent to a full one percent or more, this is still more affordable than current Universal Basic Income proposals suggest is possible.

Would the fee be progressive or regressive?

The best structure of the fee optimizes revenue while minimizing impact. The fee is charged the same way for every transaction in the entire system, all the way down the supply chains (not just the last retail sale). It applies to all financial activity equally. It discourages most of the financialization trends seen today. To make it slightly progressive, which is a divisive topic, it only applies a small increased rate to the wealthy. If an account has more zeros (a billion vs a thousand), it charges 6% more on every transaction that account is involved in. This increases the cost of earning and spending, both by the size of the balance.

How exactly can that be done?

The formula simply counts the number of zeros in each account's balance at that moment and uses that number as the fee %. The minimum fee rate is 1%. To count the zeros, it uses the math function called "logarithm" on each balance that is above the floor of 10 Deltars. The fee percentage for the transaction is the average of this result for the payor and payee account balances. For those interested, the exact formula is:

$$\text{Fee\%} = \text{average} (\text{Log10}(\max (10, \text{Balance1})), \text{Log10}(\max (10, \text{Balance2})))$$

Log examples include:

$$\log(1,000)=3$$

$\log(1,000,000,000)=9$

$\log(10)$ at the floor of 1

$\log(-10,000)$ holds it at the floor of 1

What incentives does this create?

For transactions between poor people, the minimum fee rate of 1% is always charged. The people see this as paying a total of $\frac{1}{2}\%$ on the money they earn and another $\frac{1}{2}\%$ on the money they spend. This is because they split the cost both times. For transactions between two billionaires, they are paying 9% on every transaction they do together. The only way to reduce that is to only do transactions with poor people, dropping the average to 5%. Another incentive is that larger contributions by richer accounts increases the fees, pays more into the pool for commons needs and helps the poorest the most. And the follow on effect of this is to reduce the money supply by repaying more debt, thus increasing the Deltar's value.

Is this rate affected if monetary inflation raises all prices?

The fee rate is based on the actual number size of accounts. If the value of that number changes, the rate does not follow it. This means that if the value falls, the rate is effectively charging more value against those accounts. This is a slow, gradual correction to inflation but useful when applied along with increasing credit during inequality.

Safety net are complicated?

How can one pool cover all societal safety net costs?

The fees collected from all transactions must cover all insurance premiums, all grants, a healthy universal basic income (UBI) and potentially all government costs. This isn't as expensive as it sounds because the costs are optimized for minimum cost, instant, delivered just in time, and averaged across the globe. On the other side, with these costs being universal, they reinforce the help from each other. UBI grants people the negotiating power to push for better benefits or just allow people to walk away from bad jobs. Free healthcare insurance moves much of that into the preventative category and out of the emergency category. Grants to non-profit groups no longer rely on multiple middlemen taking a cut. Governments no longer need to cover safety nets at all. Crime from UBI recipients takes a dramatic fall. Criminal and terrorist recruitment fails to persuade membership due to survival fear or monetary incentives. Even lawmakers can't compete with the masses voting with their donatable Deltars. Taken together, the required fee to cover all of this remains small.

How does the pool work?

When the fee is charged on a transaction, it is tracked and the amount is simply deleted. At midnight each day, the tally is then created inside a single, global pool and sent to 3 separate categories. One is sent to an insurance / grant function. This money pays the claims that people make for both insurance and requests for grants. The rest is divided by the number of members in the entire system with that amount sent to each member's ledger. Their ledger then splits it between UBI and an amount they can only donate to a government's official department account. The UBI becomes immediately usable and the Gov portion is sent to that account as a transaction, so it pays the fee again.

Who approves the claims?

When someone files a claim for insurance or a grant, that goes to a jury style group for approval. This once in their life, voluntary group uses a private platform to accept the task for a single shift. They see the claims arrive and make a judgement. If enough of them agree before enough deny it, it is approved and a budget is set. This automated payment schedules a certain amount to be given each day to the winner for the period of time needed. The intention is that the money should be spent immediately and not sit idle at any time. This keeps transaction velocity high so more fees can refill the pool again tomorrow. Once their shift is complete, they vote to raise or lower the % going to Social and Gov, based on their new economic awareness. This vote, however, simply informs everyone else how the jury feels, letting the whole population of votes actually change it.

How much goes where?

Initially, 75% goes to UBI, 20% to Social pool and 5% goes to Gov. This roughly matches the modified need of today, when societal changes of IOUBI are implemented. However, as the

people all begin to want either the Social or Gov pool amounts to be raised or lowered, their vote to do so can move these percentages by 0.001%/day. This means that raising Social from 20% to 21% will take 1000 days of unanimous votes by the global average of votes. The same happens for the Gov %. The UBI has a minimum limit of 50% but otherwise it is simply the leftover after the other two pools' percentages are subtracted from 100%.

Shouldn't business funding be separated from personal?

How do businesses use IOUBI?

Each human automatically gets a personal account. It functions exactly like a checking account that allows overdrafting to their full credit limit. Personal account credit limits are subject to the global economy, their regional wealth rating and their personal transaction activity. A single business account is optional to each person by simply clicking a button. This enables them to run everything a business needs in the same application. But business accounts have some differences.

The business environment.

The actual account for business use functions the same but it doesn't receive the daily dividend of UBI or Gov pool donation. It also cannot vote on the Social and Gov % rate voting. It can apply for insurance and grant claims and receive Social Credit Limit donations. Social Credit Limit (SCL) donations come from personal account holders. This is a short term increase in the business's credit limit to overdraft to, donated by people who support the business's goals. The credit gets added to their existing limit but vanishes in 25% increments each quarter of a year. To maintain an ongoing negative balance, that business must constantly re-raise new SCL from the people. With enough credit to overdraft to, businesses can access the funding they need to any level needed without courting investors, both private and public. This can scale for any size business rivalling the largest existing today.

What is SCL (social credit limit)

Each day, every personal account accumulates some SCL which they cannot use themselves. They can only give it to a business to increase their credit limit. The amount is equal to 1/365th of their total personal credit limit (PCL). If they have 36,500 of PCL, they will be credited with 100 each day. Any business asking for funding will be asking for SCL and will organize meetings for donors to gather in. Since donation group sizes are limited to a minimum of 25 people, this encourages group discussion. And since the largest donation of SCL cannot be more than 4 times the smallest one, a minimum of discussion is enforced. When a donation event occurs, the total SCL received is set to expire on specific dates. Each 3 month quarter, 25% of the SCL received is destroyed from the business account. This reduces the depth to which the business can overdraft to. If their profit has not raised their balance above the new credit limit, their spending is halted until they raise more SCL or earn more revenue. Since Government accounts act identical to business accounts, they can also raise SCL from the crowd.

Is the economy too complex?

Isn't all money the same?

IOUBI has Deltars as the units to exchange. As with every currency, they can be thought of as an asset to hold, an asset to speculate on the value of, as a utility tool for moving value around, a ledger entry for debt or as a conversion mechanism to other currencies, especially the time based ones. The IOUBI system best matches the fluid by which value moves around and Deltars can best be described as a proxy for time credits. The best way to view an economy based on IOUBI Deltars, is like the relationship between a garden and the rain. As long as there's no concern that the rain will stop, the gardener can view rainwater as abundant, free and only as useful as he has enough tomato plants to make use of. Any extra rain just becomes unneeded and possibly even harmful. During times of drought, he may need to buffer from one week to the next to cover the dry week's needs but that barrel only needs to be big enough for the expected need. If he fails to save enough, he merely connects a hose to the house.

So is it money?

Deltars satisfy many definitions of fiat and physical money but it goes a step further. By equating the purchasing standard of Deltars to the personal standard of time, it ends up functioning mostly as a helper to a system where barter is managed by time units. However, since it allows the spending to happen before the earning of that time, it doesn't really act like time banking. It's more like time credit tracking. You get the credit of time to spend just because you exist and you can spend it in any transaction with another member and then you can provide the world with enough value later to repay that time you borrowed. The last major difference is that traditionally, all money has needed some arbitrary, decreed peg to assign its value as. That decree may have been authoritarian, algorithmic or physical scarcity based. Deltars ultimately get their standard by their use among the people. When the people create it and see it following what the people felt it should adjust to, the people automatically generate trust in that value.

What parts of money are missing?

Deltars don't include the concept of interest. This may sound anti-capitalist but it's actually just expanding the origins of credit to apply to today's full range of needs. Once a person accepts the massive global trust in its value stability, it becomes the best form of debt receipt available. Because it makes interest charges economically uncompetitive, this means that the concept of investing is also a lost relic. Insurance and grants still exist but not the ability to profit from their administration. Even taxes are so much more efficient that the old and new examples can barely be compared. It would be impossible to try to model the entire current global economy in a time banking system which is why all those pieces must be substituted by the equivalent of their original function. Some people may get the new system instinctively while others, stuck in the thinking of the current system, will struggle a lot.

How does it handle growth?

Since the money supply all originates from the used credit of every individual and gets destroyed by their repayment, it scales perfectly with membership. If 1000 people use it or 10B do, each account can equally create or hoard all the Deltars they want, regardless of how many people use it. And with the subtle and direct inflation offsetting algorithms, combined with the removal of monetary inflation, the trend for prices will be to perpetually fall toward their zero marginal cost. This happens automatically, regardless of the membership size. That could result in prices of everything falling to a minimum and never changing again. The cost of living for all in the world could fall to the equivalent of 10 hours of average work even while that income is fully provided by the dividend.

Does it support increased prosperity or market growth?

The most supported type of individual work that IOUBI supports is based on the production of physical goods and services. Almost everything else is based on some aspect of financialization or interest. And since neither of those can compete with free Deltars, they will struggle to survive. In other words, IOUBI supports only prosperity and fosters the most growth to those offering the products with the most value. This replaces our current race to the bottom with a race to the top.

What level of prosperity is possible?

The current economic system is severely limited in providing the funding for most innovations and startups around the world. The entire global South is deprived of enough wealth to even afford good health and education, leaving them outside the stadium when the race begins. Other entrepreneurs have been sitting on ideas for most of their lives, hoping for a way to fund their dreams. Economists tell us that 90% of new startups fail but they fail to include that predatory and extractive funding techniques are the majority cause of that. We're then told that immigration is the big problem to fear because they will be taking our jobs. But they only came here because the funding desert at home was so bad. Whether it was the direct result of crime or just a lack of affordability, they likely would have stayed without those problems. So, if all this new innovation can be funded sustainably from all these emerging communities around the world, we should see two major trends. First is the quantity of prosperous businesses should skyrocket and second, the innovation level should soar as well. Basically, one could speed up every prediction of the future by 2 to 5 times.

Does it support renting or owning?

The housing market has been subject to numerous manipulations corrupting its market from being fair for a long time. Even the best programs to aid it have come with caveats or exclusions that have altered yet another piece, making it harder to buy homes. IOUBI offers zero interest cost loans to buy homes with no down payment, no scheduled payments, no interest cost, no places to hide secret fees and yet it still has limits designed to favor owning a maximum of 1-3 homes per family. When organized landlords or corporate landlords try to compete with this, their need for extracting profit will make them uncompetitive too. The only

part of this market that they may still compete well is when convenience and very short term occupancy place a premium on specific properties. Even then, they can still be undercut by those who don't include that special value out of a sense of community, charity, altruism or simply revolt against those entities.

What's the end game?

IOUBI is designed to constantly migrate gently toward matching the support to the people with the demand from the people. Should technology advance just a little bit, IOUBI will eventually help exactly as much as is needed. Should technology completely eliminate all human paid jobs, it will migrate toward supporting what those people want as well. In the end, if this path reaches near enough to that point, money, especially Deltars, will simply become something to use as nothing more than a way to manage production by all the robots. People won't see it as good or bad. They will always have more than they need up to the point of real world consumption limits. In other words, it will become the rain water falling on their gardens.